

Professional Summary

Exercise physiologist and computational biologist specializing in skeletal muscle physiology, 3D stochastic modeling, and calcium diffusion in sarcomeres. Extensive experience in data science applications to human performance, scholarship of teaching and learning in exercise physiology, and interdisciplinary research collaboration bridging scientific discovery with educational innovation. 1000+ citations (Google Scholar).

Address

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 <https://orcid.org/0000-0002-4253-5156>

 <https://scholar.google.com/citations?user=Qx4U24oAAAAJ&hl=en>

 <https://www.researchgate.net/profile/John-Holash>

 <https://linkedin.com/in/dr-john-holash-13082b2b>

 <https://x.com/DrHolash>

Education

2009-2017 Doctorate in Muscle Physiology

University of Calgary, Canada

Discipline: Skeletal Muscle Physiology.

Main subjects: Muscle Physiology, Computational Biology, Data Analysis, Structural Modeling, Stochastic Agent based Modelling.

Thesis: "Three dimensional stochastic computer model of the skeletal muscle half sarcomere: changes in calcium diffusion caused by the myofilament lattice."

Committee: Drs., Brian MacIntosh, Henk ter Keurs, Christian Jacob, Chris Barclay.

1997-2000 Master of Science

University of Calgary, Canada

Discipline: Exercise Physiology.

Main subjects: Cycling Power, Muscle Power.

Thesis: "Validation of single maximal effort tests for power measurement."

Committee: Drs., Brian MacIntosh, Stephen Norris, Douglas Syme.

1990-1993 Bachelor's Degree in Physical Education

University of Calgary, Canada

Main subjects: Outdoor Pursuits, Leadership in Extreme Environments.

Senior Project: "Calgary River Cleanup, Conservation of Calgary Rivers and Pathways".

Research and Professional Experience

2019- Current Professor (Teaching) - Exercise Physiology

Faculty of Kinesiology, University of Calgary

Teaching and research in exercise physiology, data science, and computational biology. Leading undergraduate and graduate instruction while supervising student research in muscle physiology and performance analysis. Granted tenure and promoted to Associate Professor effective July 1, 2025.

- 2013-2018 **Data Scientist / Systems Architect** Human Performance Laboratory, University of Calgary
Designed and developed computational solutions for research problems. Created custom software and algorithms for research equipment and specialized projects. Guest lecturer for multiple kinesiology courses.
- 2000-2013 **Senior Systems Analyst** Faculty of Kinesiology, University of Calgary
Technical lead for faculty IT infrastructure. Developed software solutions and managed research computing systems supporting faculty research operations.
- 1996-2000 **Research and Teaching Assistant** Faculty of Kinesiology, University of Calgary
Graduate teaching assistant for anatomy, physiology, and biomechanics courses. Research co-ordination and data analysis for exercise science studies.

Publications & Research

Published Journal Articles

- Wearable Technology in Orthopaedic Surgery: Applications and Future Directions**
Alexander W. Iwasyk, Sia S. Gaur, Alyssa Federico, John R. Holash, Fred Nicholls, Michael J. Monument, and Joseph K. Kendal
JBJS Reviews 13.7 (2025)
- Maximal active force in skinned muscle fibres from children with cerebral palsy**
Venus Joumaa, Faizan Syed, Jason J Howard, Gavin K Thomas, Latif Omerkhil, Sach Dabgotra, Isaac Odrigewitsch, Shuyue Liu, Robert J Holash, Timothy R Leonard, and Walter Herzog
Journal of Biomechanics 186 (2025) p. 112710. doi: 10.1016/j.jbiomech.2025.112710
- A Ramp- Versus Step-Transition to Constant-Work Rate Exercise Decreases Steady-State Oxygen Uptake**
Gabriele Marinari, Danilo Iannetta, John R. Holash, Robin Trama, Robin Faricier, Alessandro M. Zagatto, Daniel A. Keir, and Juan M. Murias
Medicine & Science in Sports & Exercise (May 2024). doi: 10.1249/MSS.0000000000003372
- Heavy-intensity priming exercise extends the Vo₂max plateau and increases peak-power output during ramp-incremental exercise**
Gabriele Marinari, Danilo Iannetta, Robert John Holash, Alessandro M. Zagatto, Daniel A. Keir, and Juan M. Murias
American Journal of Physiology-Regulatory, Integrative and Comparative Physiology 327.2 (2 Aug. 2024) R164–R172. doi: 10.1152/ajpregu.00016.2024
- Technological Breakthroughs in Sport: Current Practice and Future Potential of Artificial Intelligence, Virtual Reality, Augmented Reality, and Modern Data Visualization in Performance Analysis**
VRA Cossich, D Carlgren, RJ Holash, and L Katz
Applied Sciences 13.23 (2023) pp. 12965–12965. doi: 10.3390/app132312965
- A stochastic simulation of skeletal muscle calcium transients in a structurally realistic sarcomere model using MCell**
Robert John Holash and Brian R. MacIntosh
PLOS Computational Biology 15.3 (Mar. 2019) pp. 1–25. *Public Library of Science*, doi: 10.1371/journal.pcbi.1006712
- An innovative ergometer to measure neuromuscular fatigue immediately after cycling.**
Douglas Doyle-Baker, John Temesi, Mary E Medysky, Robert J Holash, and Guillaume Y Millet
Medicine and Science in Sports and Exercise 50 (2 Feb. 2018) pp. 375–387. doi: 10.1249/MSS.0000000000001427
- Skeletal muscle fatigue—regulation of excitation—contraction coupling to avoid metabolic catastrophe.**
Brian R MacIntosh, Robert J Holash, and Jean-Marc Renaud
Journal of Cell Science 125.9 (2012) pp. 2105–2114. *The Company of Biologists Ltd*
- A comparison of exer-gaming interfaces for use in rehabilitation programs and research.**

Kazumoto Tanaka, Jim Parker, Graham Baradoy, Dwayne Sheehan, John R Holash, and Larry Katz
Loading 6.9 (2012) pp. 69–81

Feasibility of the two-hour marathon is a burning issue.

Jared R Fletcher, Shane P Esau, R John Holash, and Brian R MacIntosh
Journal of Applied Physiology (Bethesda, Md.: 1985) 110.1 (2011) 282–discussion

Procedures for rat in situ skeletal muscle contractile properties.

Brian R MacIntosh, Shane P Esau, R John Holash, and Jared R Fletcher
Journal of Visual Experimentation 56 (2011)

Verfahren für die Ratte

Brian R MacIntosh, Shane P Esau, R John Holash, and Jared R Fletcher
(2011)

Books / Book Chapters

Technological Breakthroughs in Sport

Victor R. A. Cossich, Dave Carlgren, Robert John Holash, and Larry Katz
2024. Encyclopedia.pub

Cardiac Function in Exercise.

B R MacIntosh and R J Holash
Chapter chap. 8: Open Textbook of Exercise Physiology. 2024, pp. Pp. 1–60. Open Education Alberta

Skeletal Muscle Structure.

B R MacIntosh and R J Holash
Chapter chap. 4: Open Textbook of Exercise Physiology. 2023, pp. Pp. 1–60. Open Education Alberta

Power output and force-velocity properties of muscle.

B R MacIntosh and R J Holash
Chapter chap. 11: Biomechanics and Biology of Movement. 2000, pp. Pp. 193–210. Human Kinetics

Conference Presentations / Published Abstracts

Active properties of skinned muscle fibres from children with cerebral palsy.

Venus Joumaa, Jason Howard, Gavin Thomas, Sach Dabgotra, Christopher Roberts, Mahmoud Abusara, Shuyue Liu, R. John Holash, Shuyue Leonard, and Walter Herzog
Journal of Muscle Research and Cell Motility vol. 37 (2023). European Muscle Conference. Florence, Italy

Increased occupation of sarcomeric calcium buffers reduces required calcium release for similar troponin-c binding of subsequent activation.

Robert John Holash, Ian Smith, Walter Herzog, and Brian R MacIntosh
Journal of Muscle Research and Cell Motility vol. 37 (2017). European Muscle Conference. Montpellier, France

Effect of sarcomere length on calcium diffusion in a 3-D sarcomere model.

Robert John Holash and Brian R MacIntosh
Journal of Muscle Research and Cell Motility vol. 36 (2015). European Muscle Conference. Strasbourg, Austria

The importance of structure on: calcium release, diffusion, and binding in a spatially realistic 3-D sarcomere model.

Robert John Holash and Brian R MacIntosh
(2013). Biomedical Basis for Human Performance Across the Lifespan

3-Dimensional calcium kinetics; release, diffusion, binding, and uptake in a multicompartmental, skeletal muscle 1/2 sarcomere.

Robert John Holash and Brian R MacIntosh
Applied Physiology, Nutrition, and Metabolism vol. 37 (2012). Canadian Society of Exercise Physiology Conference, CSEP

Modelling calcium diffusion, binding, and uptake in a spatially realistic 3-dimensional sarcomere model.

Robert John Holash and Brian R MacIntosh

Journal of Muscle Research and Cell Motility vol. 33 (2012). *European Muscle Conference. Rhodes, Greece*

A comparison of exer-gaming interfaces for use in rehabilitation programs and research.

Kazumoto Tanaka, Jim Parker, Graham Baradoy, Dwayne Sheehan, John R Holash, and Larry Katz

Loading vol. 6.9 (2012). *Interactive Media Conference. Calgary, Alberta*

Micro-physiological simulation of calcium diffusion in a 3-dimensional sarcomere model.

Robert John Holash and Brian R MacIntosh

Applied Physiology, Nutrition, and Metabolism vol. 36 (2011). *Canadian Society of Exercise Physiology Conference, CSEP*

Can the second head of myosin bind to the adjacent thin filament?

Robert John Holash and Brian R MacIntosh

(2009). *Multi-scale Muscle Mechanics Conference. Woods Hole, Massachusetts*

Skeletal muscle filament spacing changes with contraction.

Robert John Holash and Brian R MacIntosh

Applied Physiology, Nutrition, and Metabolism (2009). *Canadian Society of Exercise Physiology Conference, CSEP*

Modelling calcium release in a simplified two dimensional skeletal muscle model using the agent-based system Netlogo.

Robert John Holash and Brian R MacIntosh

Applied Physiology, Nutrition, and Metabolism vol. 33 (2008). *Canadian Society of Exercise Physiology Conference, CSEP*

Validation of single maximal effort tests for peak power output.

Robert John Holash, Igor Kopecky, Krista Sevdhal, and Brian R MacIntosh

Canadian Journal of Applied Physiology vol. 25 (2000). *Canadian Society of Exercise Physiology Conference, CSEP*

Theses

Three dimensional stochastic computer model of the skeletal muscle half sarcomere: changes in calcium diffusion caused by the myofilament lattice.

Robert John Holash

Validation of single maximal effort tests for power measurement.

Robert John Holash

Masters of Science Thesis, University of Calgary

Research Supervision & Mentoring

Current Graduate Students - Committee Member

2025- **MSc. Alyssa Federico**

Medical Science

Present *Evaluating the Patient-Reported Outcome Measure Information System (PROMIS) in Sarcoma Patients*

Graduated Students - Committee Member(2021-2026)

2024-2026 **MSc. Alex Iwasky**

Medical Science

ACTIVATION: MSK Oncology| Mobility and Patient Reported Outcomes (PRO). Graduated March 2026.

2019-2026 **PhD. Jim Griffiths**

Faculty of Kinesiology

Heart Rate Novel Methods of Detection. Graduated 2026.

2021-2025 **PhD. Keenan McDougal**

Faculty of Kinesiology

Alterations in fatigue, efficiency, and pedaling mechanics during incremental and constant-load high-intensity cycling. Graduated July 29, 2025.

2022-2024 **PhD. Gabriele Marinari**

Faculty of Kinesiology

New approaches to characterize the $\dot{V}O_2$ slow component and its physiological mechanisms

2022-2024	MSc. Brynn Lindstrom <i>Does critical power underestimate thresholds in experienced exercisers</i>	Faculty of Kinesiology
2023-2024	MSc. Timi Ajayai <i>Detrended fluctuation analysis of heart rate data during constant intensity exercise</i>	Faculty of Kinesiology
2022-2024	MSc. Marissa Doroshuk <i>Novel Ovulation Research-Recruitment Methods for an App Study</i>	Faculty of Kinesiology
2022-2024	MSc. Alissa Kazakoff <i>Novel Ovulation Research-Recruitment Methods for an App</i>	Faculty of Kinesiology
2021-2022	MSc. Mary Mackie <i>The "Step-Ramp-Step" Protocol: Evaluating the Effects of a Smaller First Step Amplitude and Different Ramp Slopes to Determine the VO₂ Mean Response Time and the Expression of the VO₂ Slow Component During Ramp-Incremental Tests</i>	Faculty of Kinesiology

Graduate Student Mentoring

2021-2022	Jesse Oswald Mentoring undergraduate students in Honours Research projects	MKin Capstone Project
2021-2022	Keenan McDougal CURE Coach Course research project design in Kinesiology 375	CURE Program
2021-2022	Krystyna Sandowski CURE Coach Course research project design and Alternative Laboratory Teaching Methods in Undergraduate Kinesiology	CURE Program & MKin Capstone

Current Undergraduate Research Students (2024-2026)

2026	Biomedical Engineering-Kinesiology Summer Research Studentship - John Al-Rachid Does Menopause Accentuate Stroke-Induced Muscle Weakness in Rats?
2026	PURE Award - Faraja Herve Habiyakare Effect of Menopause and Stroke on Skeletal Muscle Sarcomeric Proteins in a Rat Model
2025	NSERC Award - Sunnie Vuong Structural Characteristics of Two Muscles from Children with Cerebral Palsy
2025	PURE Award - Abdul-Samad Ahmed Structural properties and mitochondrial function of cardiac muscle in response to a high-fat and high-sucrose diet in female rats
2024-2025	Honours 590 - Dana Litvak Cold Water Immersion for Exercise-Induced Muscle Damage Recovery: A Systematized Umbrella Review
2024-2025	Honours 590 - Sia Gaur Assessing the Impact of Exercise Duration, Intensity, and Modality on Cardiovascular Health Outcomes Using Wearable Technology: A Correlational Analysis

Former Undergraduate Research Students (2020-2024)

2024-2025	Honours 590 - Andrew Filewich Multimedia-Experiential Learning in KNES 375: Developing Demonstration Videos and Interactive Learning Modules for Physiological Assessment Techniques (MELT grant supported)
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- 2024 **NSERC Award - Arianna Hu**
Investigating the effect of flash freezing on active force production of skinned fibres
- 2024 **Summer Internship - Latif Oerkhil**
Investigating active force and sarcomeric protein content in gracilis muscle of children with cerebral palsy
- 2023-2024 **Honours 590 - Conner Hass**
Palmer cooling and the effect on metabolic work production on exercise performance
- 2023-2024 **Honours 590 - Thomas Manktelow**
X-country skiing double poling efficiency
- 2023 **NSERC Award - Birtej Mangat**
Investigating the Mechanical Properties of Cardiac Muscle in Obese Rats
- 2023 **Summer Internship - Latif Omerkhil**
Investigating active force in muscle fibres from children with cerebral palsy
- 2022-2023 **KNES 466 - Ashley Matesic**
Relative Hip Drop Timing at Different Speeds and its Relationship to Performance and Force Production in Cross-Country Skiing Double Poling
- 2022 **Faculty Research Funding - Thomas Manktelow**
Objective Analysis of Double-Pole Timing in X-Country Skiing
- 2022 **PURE Studentship - Gavin Thomas**
Active force in skeletal muscle fibres from children with cerebral palsy
- 2021-2022 **KNES 490 Honours - Allysan Lui**
Assessment of Aerobic power in Collegiate Contemporary Dance Using a High-Intensity Dance Performance Fitness Test (DAFT2)
- 2021-2022 **KNES 490 Honours - Chantal Vogel**
Effect of Zwift's Virtual Setting on Individual Outcomes and Performance in Comparison to Traditional Stationary Cycling
- 2021-2022 **KNES 490 Honours - Andreas Cordido**
Exercise Thresholds: Functional Threshold Power on an Exergaming Platform versus Power Output at the Respiratory Compensation Point
- 2021 **Biomedical Engineering - Maleeka Malik**
Investigating the changes in titin isoforms and concentration and sarcomere organization in skeletal muscles of obese rats
- 2020-2021 **KNES 490 Honours - Ashley Lorenz**
Investigating Physical Activity Intensity of Virtual Reality Exergame in Recreationally Active Young Adults
- 2020-2021 **KNES 604 - Tara McNeil**
Physiology of Swimming

Student Presentations (2025-2026)

- 2025 **BURP Summer Student Symposium** Aug 15, 2025
Investigating active force and sarcomeric protein content in gracilis muscle of children with cerebral palsy. Authors: Latif Omerkhil, Faizan Syed, Venus Joumaa, Jason Howard, Tim Leonard, John Holash, and Walter Herzog.
- 2025 **McCaig Summer Student Symposia** Aug 13, 2025
Structural Characteristics of Two Muscles from Children with Cerebral Palsy. Authors: Sunnie Vuong, Venus Joumaa, Faizan Syed, Latif Omerkhil, John Holash, and Walter Herzog.
- 2025 **McCaig Summer Student Symposia** Aug 13, 2025
STRUCTURAL PROPERTIES AND MITOCHONDRIAL FUNCTION OF CARDIAC MUSCLE IN RESPONSE TO A HIGH-FAT AND HIGH-SUCROSE DIET IN FEMALE RATS. Authors: Abdul-Samad Ahmed, Barbod Shiri, Mauricio Delgado, Nada Abu Ghazaleh, Hannah Smith, Venus Joumaa, John Holash, and Walter Herzog.
- 2025 **Transdisciplinary Child & Maternal Health Trainee Research Day** Aug 26, 2025
The Structural Characteristics of Adductor Longus and Gracilis Muscles from Children with Cerebral Palsy. Poster presentation. Sunnie Vuong.
- 2025 **26th Annual AB BME Conference** Oct 25, 2025
The Structural Characteristics of Adductor Longus and Gracilis Muscles from Children with Cerebral Palsy. 8-minute oral presentation. Sunnie Vuong.
- 2025 **HPL Seminar** Nov 20, 2025
Structural Characteristics of Adductor Longus and Gracilis from Children with Cerebral Palsy. Seminar presentation. Sunnie Vuong (alongside summer student Priyal).
- 2026 **Faculty of Kinesiology Department Symposium** April 6, 2026
Multimedia-Experiential Learning in KNES 375: Developing Demonstration Videos and Interactive Learning Modules for Physiological Assessment Techniques. Honours presentation. Andrew Filewich (MELT grant supported).
- 2026 **McCaig Summer Student Symposium** Aug 12, 2026
Does Menopause Accentuate Stroke-Induced Muscle Weakness in Rats?. Authors: John Al-Rachid, Nicole A. Conquergood, Venus Joumaa, Robert John Holash, Walter Herzog, and Stephanie A. Ross.
- 2026 **BURP Summer Student Symposium** Aug 13, 2026
Does Menopause Accentuate Stroke-Induced Muscle Weakness in Rats?. Authors: John Al-Rachid, Nicole A. Conquergood, Venus Joumaa, Robert John Holash, Walter Herzog, and Stephanie A. Ross.
- 2026 **McCaig Summer Student Symposium** Aug 12, 2026
Effect of Menopause and Stroke on Skeletal Muscle Sarcomeric Proteins in a Rat Model. Authors: Faraja Herve Habiyakare, Nicole A. Conquergood, Venus Joumaa, Robert John Holash, Walter Herzog, and Stephanie A. Ross.
- 2026 **BURP Summer Student Symposium** Aug 13, 2026
Effect of Menopause and Stroke on Skeletal Muscle Sarcomeric Proteins in a Rat Model. Authors: Faraja Herve Habiyakare, Nicole A. Conquergood, Venus Joumaa, Robert John Holash, Walter Herzog, and Stephanie A. Ross.

Grants and Awards

Major Research Grants

2025	Alberta Spine Foundation Project: <i>ACTIVATION: Activity Capture To Investigate Voluntary AcTivity in Oncology and Normal populations</i> . Co-investigator with Drs. Joe Kendal and Fred Nicholls.	Grant Value: \$75,000
2025	SPARK Alberta Project Project acceptance with Drs. Joe Kendal and Fred Nicholls to advance and commercialize the technology and research application. Chief technical designer and programmer, and researcher.	Activation-Health app
2024	Spark Grant - McCaig Institute for Bone and Joint Health Project: <i>ACTIVATION: Activity Capture To Investigate Voluntary Activity in Oncology and Normal populations</i> . Co-investigator with Dr. Joe Kendal.	Grant Value: \$20,000
2024	Science of Teaching and Learning (SOTL) Grant Project: <i>Evaluating Enhanced Learning Outcomes: Developing a Multimedia Library with Educational Videos and Interactive Tools for Laboratory Training</i> . Co-investigator with Dr. Venus Joumaa.	Grant Value: \$40,000
2021	CURE - Curriculum Based Undergraduate Research Experience Project: <i>Redevelopment of Lab component of Kinesiology 375 so that final lab experiment is student enquiry driven</i> . Taylor Institute for Teaching and Learning.	Grant Value: \$10,000
2019	Faculty of Kinesiology Startup Funds Research startup funding for new faculty position. University of Calgary.	Grant Value: \$40,000

Student Training Grants

2026	Biomedical Engineering-Kinesiology Summer Research Studentship - John Al-Rachid Grant Value: \$7,500 Project: <i>Does Menopause Accentuate Stroke-Induced Muscle Weakness in Rats?</i> .	
2026	PURE Award - Faraja Herve Habiyakare Project: <i>Effect of Menopause and Stroke on Skeletal Muscle Sarcomeric Proteins in a Rat Model</i> .	Grant Value: \$7,500
2025	NSERC Undergraduate Student Research Award - Sunnie Vuong Project: <i>Structural Characteristics of Two Muscles from Children with Cerebral Palsy</i> .	Grant Value: \$6,000
2025	PURE Award - Abdul-Samad Ahmed Project: <i>Structural properties and mitochondrial function of cardiac muscle in response to a high-fat and high-sucrose diet in female rats</i> .	Grant Value: \$7,000
2024	NSERC Undergraduate Student Research Award - Arianna Hu Project: <i>Investigating the effects of flash freezing on active force production of skinned muscle fibres</i> .	Grant Value: \$6,000
2023	NSERC Undergraduate Student Research Award - Birtej Mangat Project: <i>Investigating the Mechanical Properties of Cardiac Muscle in Obese Rats</i> .	Grant Value: \$6,000
2022	PURE Studentship Grant - Gavin Thomas Project: <i>Investigating active force in skeletal muscle fibres from children with cerebral palsy</i> .	Grant Value: \$7,000
2022	Faculty of Kinesiology Undergraduate Research Scholarship - Thomas Manktelow Value: \$7,000 Project: <i>Objective Analysis of Double-Pole Timing in X-Country Skiing</i> .	Grant

Teaching and Recognition Awards

2023	Faculty of Kinesiology COVID Outstanding Achievement Award Recognition for exceptional contributions during the COVID-19 pandemic period.	University of Calgary
2021	BME/Faculty of Science Summer Student Best Presentation Award For Maleeka Malik: <i>Titin isoform changes in an obesity feeding rat model.</i>	University of Calgary
2013	Young Investigator Award Best Presentation: <i>The importance of structure on calcium release, diffusion, and binding in a spatially realistic 3-D Sarcomere Model.</i> Bio-medical Basis for Human Performance Across the Lifespan.	University of Calgary
2013	Outstanding Leadership (Staff) Award University of Calgary recognition for exceptional leadership contributions.	Roger Jackson Centre for Health and Wellness

Research Travel and Development Grants

2012	Research Travel Grant Support for conference presentation and research collaboration.	Faculty of Graduate Studies, University of Calgary
2011	Excellence in Research Grant Graduate student research support funding.	Faculty of Graduate Studies, University of Calgary

Historical Awards and Recognition

1998	Alberta Sports Research Grant Development of electronic bike ergometer for sports performance research.	Government of Alberta
1996	Alberta Parks and Recreation Grant Measuring muscle tone in children with Downs Syndrome.	University of Calgary
1996	You Make a Difference Award Recognition for blind bowling program development and leadership.	Canadian National Institute for the Blind
1994	Clean World Award For accomplishments running the Calgary River Clean-up 1994.	International Association for Environmental Urban Living
1993	Mayor's Environmental Stewardship Award Presented by Mayor Al Duer for organizing and running the Calgary River Clean-up 1993.	City of Calgary

Grant Summary Total research funding secured: \$185,000+ | Student training awards supervised: \$54,000 | Teaching innovation grants: \$50,000

Teaching & Courses Taught

Undergraduate Courses

2019-2026	KNES 213 - Introduction to Kinesiology Fall/Winter/Spring terms 11+ sections Recent enrollments: 146 (Fall 2025), 150 (Winter 2026) An introduction to research in kinesiology with an emphasis on understanding the research process, including basic statistical knowledge, and its relationship to critical thinking. Practical application of concepts through direct involvement in individual and group projects.	
2020-2025	KNES 375 - Tests and Measurements Fall/Winter terms 5+ sections Recent enrollment: 86 (Fall 2025) Establishment of tests, criteria for selection of tests, measurement devices used to evaluate physiological status, human growth, and skill levels in physical activity programs.	

2010-2026 **KNES 381 - Computer Applications** Fall/Winter terms | 4+ sections | Recent enrollments: 25 (Winter 2025, Winter 2026)

An introduction to the use of the computer in kinesiology which involves hands-on experiences with selected software packages.

2020-2021 **KNES 355 - Human Growth and Development** Winter terms | 2 sections | Avg: 250 students/section

The physiological, anatomical, emotional and social changes in human growth and development, with a view to the planning and selection of appropriate programs in physical education, sport, and dance.

2022 **KNES 475 - Physiological Bases of Athletic Performance** Winter term | 1 section | 40 students

The physiological factors and principles of training affecting performance will be reviewed and challenged on the scientific basis of experimental evidence.

Graduate Courses

2020-2024 **KNES 606 - Applied Exercise Physiology Practicum** Fall/Winter terms | 8 sections | Avg: 14 students/section

Comprehensive graduate-level laboratory course covering advanced exercise testing and research methodology.

2025-2026 **KNES 613/614 - Applied Exercise Physiology Practicum** Fall/Winter terms | Recent enrollment: 15 (Fall 2025, Winter 2026)

Comprehensive graduate-level laboratory course covering advanced exercise testing and research methodology.

Honours Students and Self-Directed Studies

2020-2025 **Honours Research Projects** 15+ students supervised

KNES 490/590 Honours projects, KNES 466 Advanced Biomechanics projects, and KNES 604 Advanced Studies. Covering diverse research areas including VR fitness, dance performance, skiing biomechanics, swimming physiology, and exercise technology applications.

Teaching Summary Course portfolio: 6 different courses | Total sections taught: 31+ | Students instructed: >2,600 | Teaching innovations: Flipped classroom methodology, student-driven laboratory experiments, VR/technology integration | Recognition: Featured in university teaching showcase, CURE grant recipient, COVID Outstanding Achievement Award

Patents, Licenses and Innovations

2025-2026 **Activation-Health** Apple App Store
App development and release. <https://healthtracker.ucalgary.ca/>

2025-2026 **Sauna Timer** Apple App Store
App development and release.

2025-2026 **Plunge Timer** Apple App Store
App development and release.

Service and Professional Activities

Research Institute Memberships

2024-
Current **Alberta Children's Hospital Research Institute** University of Calgary
Alberta Health Services and the Alberta Children's Hospital Foundation

2024- Current	McCaig Institute for Bone and Joint Health Associate Member, Cumming School of Medicine	University of Calgary
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Faculty and University Committee Service

2024	Search Committee: Exercise Physiology Position University of Calgary	Faculty of Kinesiology
2022- Current	Teaching and Learning Committee University of Calgary	Faculty of Kinesiology
2021- Current	Ethics Committee for Human Studies Faculty representative, University of Calgary	Faculty of Kinesiology
2019- Current	Learning Technologies Advisory Committee Faculty of Kinesiology representative	Taylor Institute for Teaching and Learning
2021-2022	National Survey of Student Engagement University of Calgary	Faculty of Kinesiology
2021	Search Committee: Human Growth and Development Position University of Calgary	Faculty of Kinesiology
2020	Student Orientation Online Technologies Special Committee University of Calgary	Central Orientation Committee
2019-2021	New Student Orientation University of Calgary	Faculty of Kinesiology
2019-2021	Student Orientation - Faculty Advisor Panel University of Calgary	Faculty of Kinesiology
2019	YUJA Video Evaluation Group Faculty of Kinesiology representative	Taylor Institute for Teaching and Learning

External Academic Service

2020- Present	Thesis Examiner - PhD Candidacy Exams Thomas Tripp (2023), Cody van Rassel (2022), Nada Abughazaleh - Biomedical Engineering (2021), Calaine Inglis (2020)	Various Faculties
2020- Present	Thesis Neutral Chair 15+ doctoral and master's thesis defenses across Kinesiology and Biomedical Engineering	Various Faculties
2018-2019	Invited Reviewer Calgary, Alberta	International Society of Biomechanics

Conference Organization and Support

2019	Team Captain, Speaker Ready Room International Society of Biomechanics Conference, Calgary	ISB/ASB Calgary
2019	Course Maintenance World Cup at Canmore Nordic Centre	International Biathlon Union
2006-2007	Technology Coordinator Canadian Society for Exercise Physiology, Banff	CSEP Conference

2002	Presentation Assistant Calgary, Alberta	World Congress of Biomechanics
1999-2000	Technology Coordinator Canadian Society for Exercise Physiology, Canmore	CSEP Conference
1999	Presentation Assistant Calgary, Alberta	International Society of Biomechanics
1995-1996	Technology Director Calgary, Alberta	Special Olympics Canada Winter Games

Professional Memberships

1997- Current	Canadian Society for Exercise Physiology 27+ years membership	CSEP
2010- Current	European Muscle Physiology Society 14+ years membership	EMC
2010- Current	Canadian High Performance Computing Society 14+ years membership	HPC
2025- Current	European Congress of Sports Science New member	ECSS

Volunteer and Community Service

2003-2016	Board Member CIRA - Technology governance and policy	Canadian Internet Registration Authority
1995-1996	Technology Director & Program Coordinator Special Olympics Canada Winter Games and swimming programs	Special Olympics Calgary
1995-1996	Program Facilitator Preparation for Re-entry into Education Program (PREP)	Grace Hospital
1993-1996	Environmental Director Environmental stewardship and outdoor education	Calgary Canoe Club & Calgary Area Outdoor Council
1993-1996	Environmental Advisor City of Calgary environmental initiatives	Mayor's Environmental Committee
1992-1995	Emergency Room Support Volunteer healthcare support	Calgary General and Rocky View Hospitals

Service Summary Committee service: 15+ university committees | External service: 20+ thesis examinations | Conference organization: 8+ major conferences | Professional memberships: 25+ years combined | Community leadership: 20+ years volunteer service

Media & Public Engagement

2026	UToday News Article: ACTIVATION Health: Made-at-UCalgary app aims to change how recovery is measured https://www.ucalgary.ca/news/activation-health-made-ucalgary-app-aims-change-how-recovery-measured	UCalgary News Sept. 17, 2026
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2025	Exploring the AI Running Revolution https://open.spotify.com/episode/0yN8RkAbjMh3j2qS2ljOW8	Run to the Top Podcast The ultimate running podcast
2025	Can A Weighted Vest Amp Up Your Daily Walk? https://www.canadianliving.com/health-fitness/article/can-a-weighted-vest-amp-up-your-daily-walk	Canadian Living: Sara Romano
2025	How healthy is the polar plunge? https://www.cbc.ca/listen/live-radio/1-91-the-early-edition/clip/16189894-how-healthy-polar-plunge	CBC Radio One - The Early Edition with Stephen Quinn Dec 31, 2025
2025	Cold plunge for the new year	CBC Radio One Vancouver Dec 31, 2025 7:45am
2025	The science behind cold plunging https://www.cbc.ca/listen/live-radio/1-91-the-early-edition/clip/16189894-how-healthy-polar-plunge	Connect with Sarah Crosbie (AM 770 CHQR, QR Calgary) Dec 18, 2025 3:45pm
2024	Technology is revolutionizing sport performance. https://www.abc.net.au/listen/programs/futuretense/technology-future-of-sport-cheating-advancement-performance-peak/104040538	Future Tense Podcast
2024	Calgary Herald: How to cold down when the heat is on. https://calgaryherald.com/health/diet-fitness/how-to-cool-down-when-the-heat-is-on	Calgary Herald
2024	N.S. woman says ocean plunges offering her health benefits https://globalnews.ca/video/10233505/n-s-woman-says-ocean-plunges-offering-her-health-benefits/	Global News
2024	Cold Plunges for the new year January 8th	CBC -Edmonton Live
2024	West coast cold plunges January 5th	CKNW 980 Vancouver with Scott Shantz
2024	Cold Plunge fad or fact January 4th	CHEB -Edmonton Live
2024	Cold plunges are all the rage. But what does the science say? https://www.cbc.ca/news/canada/cold-plunges-1.7072906	CBC news
2023	Innovator/Educator –Dr. John Holash https://youtu.be/ht2z2Jpy1Bc?si=M88XOgBj6ROXxCmF	Taylor Institute for Teaching and Learning Video series
2023	Are you ready to take the plunge? Experts weigh in on cold plunge benefits https://globalnews.ca/the-curator/10042896/cold-plunge-benefits/	Global News
2023	Best of Health Magazine:The Many Health Benefits of Nordic Skiing https://www.besthealthmag.ca/article/nordic-skiing-cross-country-skiing?_cmp=stf	readers digest
2023	Why some people are taking a wintry dip from the banks of the Bow River https://t.co/yGpoTu8sB5	CBC News
2022	UToday News Article: Course revamp is a hit with kinesiology students when they create their own fitness tests https://news.ucalgary.ca/news/course-revamp-hit-kinesiology-students-when-they-create-their-own-fitness-tests	UToday

2022	Council on Undergraduate Research: Course revamp is a hit with kinesiology students when they create their own fitness tests https://www.cur.org/course-revamp-is-a-hit-with-kinesiology-students-when-they-create-their-own-fitness-tests/	CUR.org
2022	KQ Education Group: Course revamp is a hit with kinesiology students when they create their own fitness tests https://kqeducationgroup.com/course-revamp-is-a-hit-with-kinesiology-students-when-they-create-their-own-fitness-tests-news/	KQ Education Group
2021	Spotify Podcast COVID Coffee Chats @ Ucalgary Episode 8: Creating a flipped Classroom with John Holash https://open.spotify.com/episode/1yF8F4Zn62JHdBuZ1LB6q?si=0ead0ddf70f24661	Spotify Podcast
2021	Calgary Journal by Lee Reed: Connection between Mental Health and Exercise. Interview with Dr. John Holash	Calgary Journal
2021	UToday News Article: HealthyU team creates accessible, cost-friendly workouts catered to students' busy lives https://www.ucalgary.ca/news/healthyu-team-creates-accessible-cost-friendly-workouts-catered-students-busy-lives	UToday

Training & Learning

2025	European College of Sports Science Congress July 1-4, 2025	Rimini, Italy
2024	Faculty of Kinesiology Teaching and Learning retreat	Alt Hotel
2022-24	Kaggle.com Computer programming and Data visualization programming	online
2022	TI 0746-002 Developing Your Teaching Dossier for Tenure and/or Promotion	Taylor Institute
2022	TI 0913-003 Creating a Flipped Lesson	Taylor Institute
2022	TI 0765-004 Intentional D2L Course Shell Design	Taylor Institute
2022	TI 0795-003 Online Student Assessment	Taylor Institute
2022	TI 0783-001 Undergraduate Research and Experiential Learning: Focusing Strategies for Courses & Programs	Taylor Institute
2022	Academic Integrity in Online courses: Adapting during COVID (March 25)	Taylor Institute
2021	Learning to teach online	Linda Learning
2021	Data science essentials with R	Linda Learning
2021	Creating fun and Engaging Video Training: The Why	Linda Learning
2022	Learning Git and GitHub	Linda Learning
2022	Web Scraping in Python	Linda Learning
2022	Using Python with Excel	Linda Learning

2022	Excel Advanced formulas and Functions	Linda Learning
2020	Putting your course online (March 23)	Taylor Institute
2020	Increasing engagement with eLearning programs	Linda Learning
2020	eLearning essentials: Visual design	Linda Learning
2020	eLearning essentials: Instructional design	Linda Learning
2020	Developing and delivering online courses	Taylor Institute
2019	Teaching Days	Taylor Institute
2019	Data Science with Python	Linda Learning
2018	Spill Response Training	Online, UofC
2018	Bio-Safety Training	Online, UofC
2017	Chematix / Lab Manager	Chematix, UofC
2007	Animal Care and Handling	Online, UofC
2018	Occupational Health and Safety Orientation	Online, UofC
2018	Workplace Inspections Training	Online, UofC
2016	WHMIS 2015	Online, UofC
2016	Bio-Safety Program Training	Online, UofC
2016	Hazard Assessment Training	Online, UofC

Certifications

2020	RSO- Range Safety officer	Canmore Nordic Centre
2016	FIS X-Country Ski Official cross country skiing international federation of sport, officials course	Canmore Nordic Centre
2013	ADI Instruments System Management and Teaching basics ADI System Management and Physiology Instruction modules	University of Saskatoon
2008	ITIL Intermediate Level V3 certification Standards for Computer Support	University of Calgary
2005	Management Training Franklin Covey Leadership	Franklin Covey Leadership
2004	Microsoft Certified System Architect MCSA	Continuing Education, UofC
2001	Microsoft Certified Database Professional MCDP	Continuing Education, UofC

1998	Microsoft Certified Professional MCP	Continuing Education, UofC
1994	Canadian Association of Alpine Ski Instructors CSIA Level I	Canada Olympic Park, WinSport
1994	Canadian Association of Nordic Ski Instructors CASI Level I	Canada Olympic Park, WinSport
1992	Canadian Recreational Canoe Association CRCA Level V	Calgary Canoe Club

September 25, 2026

John Holash